SHIVAM KAMBOJ

Quantum Control · Characterization · Device Modeling

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Ph.D. candidate in Physics, UC Merced (expected Fall 2026). At LLNL, wrote pulse-level control and linear-inversion process-tomography software for superconducting qudits. Thesis research models ultrastrong light–matter coupling in the quantum Rabi dimer, including COMSOL models of a proposed superconducting implementation. Also model magnon–photon systems with an experimental group.

TECHNICAL SKILLS

- **Control and characterization:** pulse-level control of superconducting qudits; optimal-control gate sequences; linear-inversion quantum process tomography (pseudo-inverse reconstruction); Ramsey and T_1/T_2 fitting.
- **Device and noise modeling:** COMSOL RF and AC/DC; capacitance, impedance, eigenmode, and inductive-participation extraction; circuit quantization; YIG–cavity magnon–photon models; $1/f$ noise models.
- **Software:** Python, NumPy, SciPy, QuTiP, Qiskit, Stim, PyMatching, Matplotlib; SLURM, Linux, Git, GitHub Actions; `pytest`, `mypy`, `ruff`; Flask, Gemini API, Google Cloud.

EXPERIENCE

Lawrence Livermore National Laboratory — *Physics Graduate Intern, Quantum Coherent Device Physics Group* **May – Aug 2024**

- Wrote Python software for pulse-level control of multilevel superconducting qudits (transitions up to $|3\rangle$) and evaluated optimal-control gate sequences.
- Built a generalized linear-inversion process-tomography pipeline (pseudo-inverse reconstruction) that turns measured data into process maps used to characterize the gates.
- Automated Ramsey and T_1/T_2 fits to give quantitative feedback during control experiments.

Tian Group, University of California, Merced — *Graduate Student Researcher (advisor: Prof. Lin Tian)* **May 2023 – Present**

- Build COMSOL electromagnetic models of a proposed superconducting implementation of the quantum Rabi dimer; extract capacitances, mode frequencies, impedances, and inductive participation to check which couplings the layout can realize.
- Simulate dissipative dynamics (photon loss, entanglement) with Lindblad master equations in QuTiP and sparse exact diagonalization; run checkpointed parameter studies on SLURM clusters.
- Combine normal-mode analysis and parity-resolved diagonalization to separate true from avoided level crossings and characterize a candidate cat-state doublet.
- Proposed extension (independent): test whether this doublet can support a noise-biased bosonic qubit.

Sharping Group (UC Merced) and Hurst Group (SJSU / UC Merced) — *Research Collaborator; Graduate Researcher* **Jun 2021 – Present**

- Develop Hamiltonian models of coupled YIG–cavity (magnon–photon) systems with inter-cavity coupling and parametric driving, working with an experimental quantum-optics group.
- Derived dispersion relations and numerical solvers for 2D spin-torque oscillator arrays and studied how edge modes respond to disorder (first-author paper, *Phys. Rev. B*).

SOFTWARE

`nonmarkov_qec` — *open-source Python package for surface-code simulation under correlated noise; released on PyPI* **2026**

- Combines Ornstein–Uhlenbeck and $1/f$ dephasing models, Stim circuit sampling, PyMatching decoding, and bootstrap Monte Carlo analysis to measure how temporally correlated noise affects surface-code performance.
- 74 automated tests; strict type checking, linting, and CI with automated releases.

Physics Concept Analyzer Tool (PCAT) — *Python, Flask, Gemini API, Google Cloud* **2025**

- Built and deployed a web application on Google App Engine that guides physics students through model construction before calculation; piloted in UC Merced physics courses.

EDUCATION, PUBLICATIONS AND AWARDS

Ph.D. in Physics, UC Merced (expected Fall 2026; advisor: Prof. Lin Tian) · **M.S. in Physics**, San Jose State University (2022)

1. S. Kamboj, R. A. Duine, B. Flebus, and H. M. Hurst, "Oscillatory edge modes in two-dimensional spin-torque oscillator arrays," *Phys. Rev. B* **109**, 094436 (2024).

2. S. Kamboj, J. Beurkens, J. Sharping, and L. Tian, "Superconducting realization of the quantum Rabi dimer" (in preparation).

3. S. Kamboj, Y. Cho, and Y. J. Rosen, "Improving fidelity of quantum gates using optimal control," poster, LLNL PLS Student Poster Symposium (2024).

Awards: Outstanding Graduate Teaching Award, UC Merced (2026); Outstanding Teaching Assistant Award, SJSU (2022); Physics Graduate Group Summer Fellowship (2025, 2026).